08 - Configure Voice Controls

In this exercise, you'll enhance your multi-modal chatbot by adding voice input and output capabilities. Users will be able to speak to your chatbot instead of typing, and receive spoken responses, creating a truly conversational AI experience.

What You'll Learn

By completing this exercise, you will:

- **Implement** speech-to-text functionality for voice input
- **Add** text-to-speech capabilities for voice output
- **Create** a hands-free conversational experience
- **Experience** browser-based voice API integration with Kiro

Step 1: Add Voice Input

Start by adding the ability for users to speak to your chatbot:

```
Now I would like to add the ability to talk to the chatbot, instead of typing.
Please implement voice input functionality so users can speak their messages.
```

Browser Compatibility
Voice features may only be available in certain browsers like Chrome. Kiro will implement the best available solution for cross-browser compatibility.

Step 2: Add Voice Output

Once voice input is working, enhance the experience with voice responses:

```
Now I would like to add the ability for the bot to talk back in voice as well. Please implement text-to-speech so the chatbot can speak its responses.
```

What Kiro Will Update

Your voice enhancement will include:

- **Voice Input**: Speech-to-text functionality using browser APIs
- **Voice Output**: Text-to-speech for chatbot responses
- **UI Controls**: Voice recording and playback controls
- **Browser Integration**: Web Speech API implementation
- **Fallback Options**: Graceful degradation for unsupported browsers

Complete Voice Experience
Your chatbot will now support both listening to users and speaking responses, creating a natural conversational interface.

Step 3: Test Voice Functionality

Please test both voice input and output features. Make sure you can speak to the chatbot and hear responses back.

Expected Behavior

Your enhanced chatbot should demonstrate:

▶ **Voice Input Example**

▶ **Voice Output Example**

Step 4: Update Documentation

Since you've built an impressive multi-modal AI chatbot, update your project documentation:

```
Can you generate or update the readme file with the details of what we have done? Include all the features we've implemented throughout the workshop.
```

Success Criteria

After completing this exercise, you should have:

- ✅ **Voice Input**: Users can speak messages to the chatbot
- ✅ **Voice Output**: Chatbot responds with spoken audio
- ✅ **Browser Integration**: Proper use of Web Speech APIs
- ✅ **UI Controls**: Intuitive voice recording and playback interface
- ✅ **Updated Documentation**: Comprehensive README reflecting all features

Voice-Enabled AI Complete
Your chatbot now offers a complete hands-free conversational experience with voice input, text/image generation, and voice output!

Troubleshooting

Common issues you may encounter with voice features:

▶ **Browser Compatibility Issues**

▶ **Microphone Access Denied**

▶ **Voice Recognition Accuracy**

▶ **Audio Playback Issues**

Architecture Overview

▶ **View Complete Voice-Enabled Architecture**

This exercise completes your transformation from a simple text chatbot to a sophisticated multi-modal AI assistant with voice, text, and image capabilities.

Key Takeaways

This exercise demonstrated advanced browser integration and voice technology concepts:

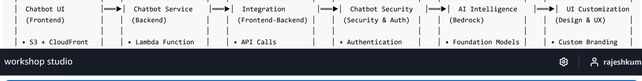
Voice Technology | User Experience | Technical Integration

- **Speech Recognition**: Implemented browser-based speech-to-text conversion
- **Text-to-Speech**: Added natural voice output for chatbot responses
- **Browser APIs**: Utilized Web Speech API for voice functionality
- **Real-time Processing**: Handled live audio input and output streams
- **Cross-Browser Support**: Implemented compatibility across different browsers

Next Steps

Your chatbot now has comprehensive voice capabilities, making it a truly hands-free AI assistant. The next exercise will add user authentication to make your application enterprise-ready with secure user management.

Workshop Progress



Let's Implement User Registration and Authentication

© 2008 - 2026, Amazon Web Services, Inc. or its affiliates. All rights reserved. Privacy policy Terms of use Cookie preferences